

- 2 Most Bengal tigers are orange with black stripes but there is a very small number of Bengal tigers that are white with black stripes.

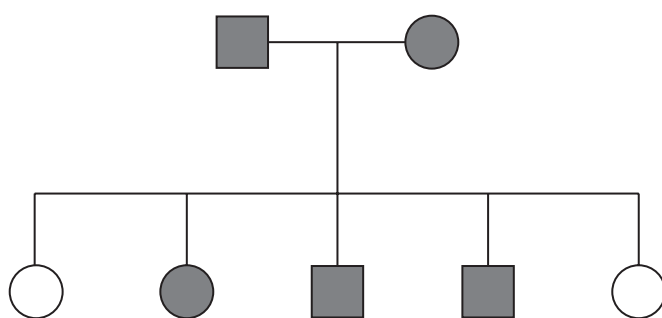
The photograph shows a white Bengal tiger with black stripes.



(Source: Caroline Wilcox)

White tiger offspring are produced by two Bengal tigers that each carry at least one recessive allele for a gene affecting coat colour.

The pedigree diagram shows the phenotypes in one family of tigers, bred in captivity.



Key

-  female orange tiger
-  female white tiger
-  male orange tiger
-  male white tiger

- (a) The phenotype is affected by the genotype.

State what is meant by the term **genotype**.

(1)

.....

.....

.....



(b) State the probability that the next tiger born to these two parents will be female.

(1)

(c) Determine the expected phenotypic ratio of orange tigers to white tigers born to the parents shown in this pedigree diagram.

Use a genetic diagram to support your answer.

(3)

Answer

(d) The incidence of white tigers in the wild is 1 in 10 000 Bengal tigers.

There are approximately 6000 Bengal tigers in captivity, 200 of which are white.

Calculate the incidence of white tigers in captivity.

(1)

Answer

(Total for Question 2 = 6 marks)

